Question 4

- The best model :

$$y_t = \alpha_0 + \alpha_1 y_{t-1} + \beta^z z_t + \beta^p p_t + \gamma_0^p \Delta p_t + u_t$$

$$y_t = \alpha_0 + \alpha_1 y_{t-1} + \beta^z z_t + \beta^p p_t + \gamma_0^p p_t - \gamma_0^p p_{t-1} + u_t$$

$$(1 - \alpha_1 L) y_t = \alpha_0 + \beta^z z_t + \left(\beta^p + \gamma_0^p - \gamma_0^p L\right) p_t + u_t$$

$$y_t = \frac{\alpha_0}{(1-\alpha_1 L)} + \frac{\beta^z}{(1-\alpha_1 L)} z_t + \frac{\left(\beta^p + \gamma_0^p - \gamma_0^p L\right)}{(1-\alpha_1 L)} p_t + \frac{1}{(1-\alpha_1 L)} u$$

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$$y_t = \alpha_0\left(1+\alpha_1 L+\alpha_1^2 L^2 + +\alpha_1^3 L^3 + \ldots\right) + \beta^z\left(1+\alpha_1 L+\alpha_1^2 L^2 + \alpha_1^3 L^3 + \ldots\right) z_t +$$
$$+\left(\beta^p + \gamma_0^p - \gamma_0^p L\right)\left(1+\alpha_1 L+\alpha_1^2 L^2 + \alpha_1^3 L^3 + \ldots\right) p_t + \left(1+\alpha_1 L+\alpha_1^2 L^2 + \ldots\right) u_t$$

Taking derivatives

$$\frac{\partial y_t}{\partial p_t} = \delta_0 = \beta^p + \gamma_0^p$$

$$\frac{\partial y_{t+1}}{\partial p_t} = \delta_1 = -\gamma_0^p + \alpha_1\left(\beta^p + \gamma_0^p\right) = -\gamma_0^p + \alpha_1 \delta_0$$

……..

$$\frac{\partial y_{t+1}}{\partial p_t} = \delta_j = \alpha_1 \delta_{j-1}$$

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$$\delta_j = \begin{cases} \beta^p + \gamma_0^p & j = 0 \\ -\gamma_0^p + \alpha_1 \delta_0 & j = 1 \\ \alpha_1 \delta_{j-1} & j \geq 2 \end{cases}$$

Impulse – response: $\Delta(j) = \sum_{i=1}^{j} \delta_i$

We do the same in order to obtain the impulse-response to the change in income.

$$\lambda_j = \begin{cases} \beta^z & j = 0 \\ \alpha_1 \lambda_{j-1} & j \geq 1 \end{cases}$$

Impulse – response: $\Delta(j) = \sum_{i=1}^{j} \lambda_i$

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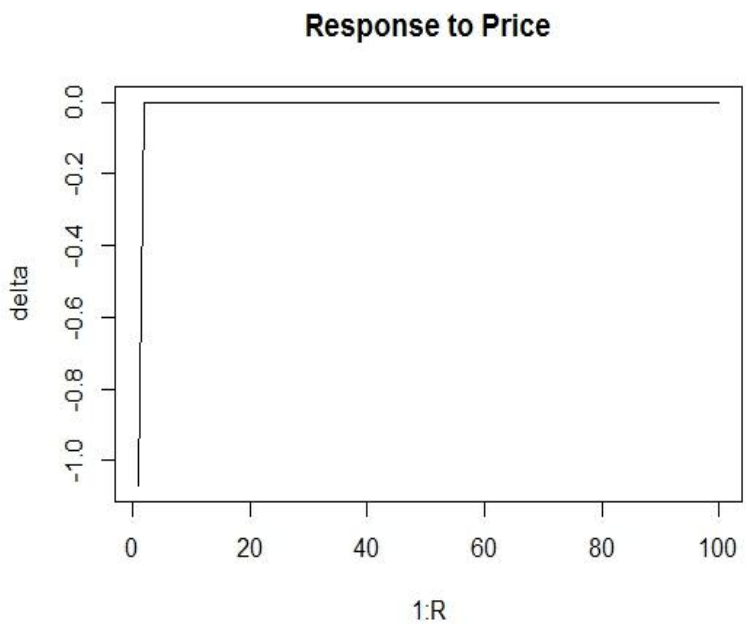

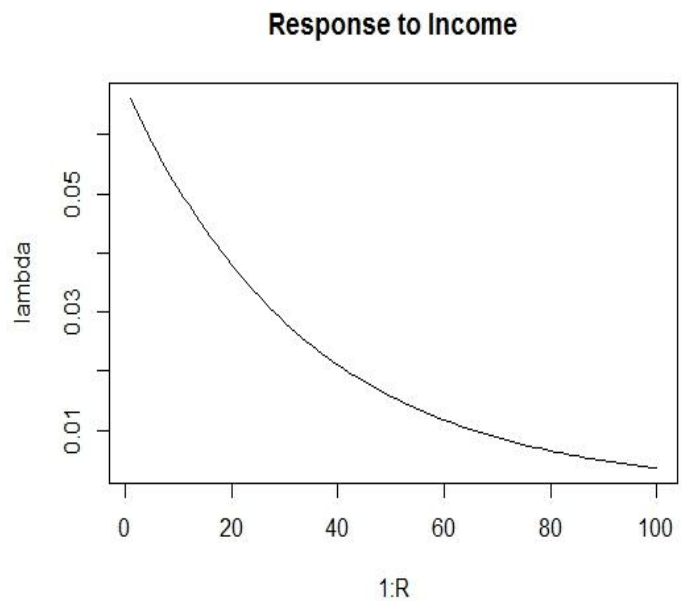

We can see that one time shock in income will have an effect in demand on gas that vanishes only slowly . The effect on price vanishes almost completely after one period.

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Model in error-correction form:

$$y_t = \alpha_0 + \alpha_1 y_{t-1} + \beta^z z_t + \beta^p p_t + \gamma_0^p \Delta p_t + u_t$$

Subtract y_{t-1} from both sides, then add and subtract $\beta^z z_{t-1}$ and finally add and subtract $(\beta^p + \gamma^p)p_{t-1}$

$$\Delta y_t = \beta^z \Delta z_t + (\beta^p + \gamma^p)\Delta p_t + (\alpha_1 - 1)\left[y_{t-1} - \frac{\alpha_0}{1-\alpha_1} - \frac{\beta^z}{1-\alpha_1} z_{t-1} - \frac{\beta^p}{1-\alpha_1} p_{t-1}\right] + \mu_t$$

We can see that the change in gas demand is decompose into two terms: the change due to the current change in the prices and income and one part is the partial correction from being out of equilibrium in previous period.